Notes: Griffin, Kruger, & Mahajan (JFE, 2026)

Did Pandemic Relief Fraud Inflate House Prices?

Contents

1 Big picture

- **Question.** Did fraud in COVID relief programs — especially fraudulent Paycheck Protection Program (PPP) loans — create a local demand shock large enough to raise house prices, stimulate consumer spending, and affect local inflation?
- **Setting.** The paper studies the United States during and after COVID, mainly from 2020 to 2023. The core spatial unit is the ZIP code, with county fixed effects to compare high-fraud and low-fraud ZIP codes inside the same local macro environment.
- **Main empirical object.** The treatment is suspicious PPP lending per capita, measured using loan-level fraud flags developed in Griffin, Kruger, and Mahajan (2023). The paper also uses broader fraud proxies based on excess loans relative to establishments and high similarity across loans.
- **Core mechanism.** Legitimate relief was intended to replace lost income or lost business revenue. Fraud, by contrast, created a windfall. If recipients spent that windfall quickly, it could increase demand for durable and immovable goods — especially housing — and temporarily distort prices.
- **Main micro result.** Individuals who received flagged PPP loans were more likely to buy homes than recipients of non-flagged loans, and were more likely to move than similar non-recipients.
- **Main housing result.** House prices in ZIP codes in the top decile of suspicious lending per capita rose 5.8 percentage points more than in ZIP codes in the bottom decile within the same county. In the paper's baseline standardized regressions, a one-standard-deviation rise in flagged loans per capita predicts 2.11 ppt higher house-price growth from January 2020 to December 2021.
- **Main credibility result.** The effect survives synthetic-control comparisons, many alternative controls, and an IV strategy that instruments local fraud with fraud in geographically distant but socially connected ZIP codes. The IV estimate is even larger: about 3.47 ppt for a one-standard-deviation increase in fraud.
- **Demand-and-reversal result.** High-fraud areas show more fast sales, more purchases above list price, more views per property, and lower inventory starting in mid-2020. Those demand metrics reverse in 2022–23, and high-fraud ZIP codes subsequently experience lower house-price growth, reversing about 35% to 41% of the initial fraud effect.
- **Other spending result.** Fraud-heavy areas also saw stronger vehicle purchases, higher Mastercard consumer spending, more visits to many kinds of retail locations, and higher regional inflation — especially housing inflation.
- **Why the paper matters.** The paper's contribution is broader than "fraud happened." It shows that fraud can create local asset-price and spending distortions that are not captured by the direct fiscal cost alone. In the post-COVID housing boom, pandemic fraud appears to be one of the quantitatively largest and most robust explanatory factors.

2 Data and measurement (key objects)

2.1 Suspicious-lending measures and why they are central

- **Loan-level source.** The backbone is the SBA's loan-level PPP data, covering 11,469,801 loans originated between April 3, 2020 and June 30, 2021, with total value of about $793 billion.
- **Primary fraud flags.** The paper uses the suspicious-loan indicators developed in Griffin et al. (2023):
  - loans to non-registered businesses,
  - multiple loans at the same residential address,
  - implausibly high implied compensation relative to local industry benchmarks,
  - and major inconsistencies between jobs reported to PPP and jobs reported to the contemporaneous EIDL Advance program.
- **Additional fraud proxies.** The paper also uses:
  - *High Loan-to-Establishment Per Capita*: county-industry pairs with more than twice as many PPP loans as Census establishments,
  - *High Similarity Per Capita*: county-lender cells with unusually similar loan amount, jobs, and industry patterns,
  - *Flagged Composite Per Capita*: a broader composite combining the flagged-loan measure with the two additional proxies above.
- **Standardization.** Most key regressors are standardized to have mean zero and standard deviation one, using population-weighted standard deviations, so coefficients can be read as effects of a one-standard-deviation change in local fraud.
- **Economic idea.** These fraud measures matter because legitimate PPP lending should mostly offset lost revenues, whereas fraudulent funds are much closer to new wealth and can therefore create excess local demand.

2.2 Housing-market, property, and address data

- **ZIP-code house prices.** The main house-price outcome is Zillow's House Value Index (ZHVI), at monthly ZIP-code frequency. After restrictions, the main sample contains 18,761 ZIP codes representing about 93% of the U.S. population.
- **Alternative housing data.** Redfin and Realtor.com provide additional market-tightness measures: share of listings sold within two weeks, share sold above list, views per property, and inventory.
- **Individual house purchases.** The paper links individual PPP borrowers to PropertyRadar and LexisNexis property records. The PropertyRadar sample contains 250,000 random individual PPP borrowers; the LexisNexis sample contains 150,000 borrowers and provides an independent validation exercise.
- **Moving behavior.** Verisk address-history data allow the authors to compare recipients of suspicious loans with otherwise similar non-recipients in the same census block group. Melissa change-of-address data provide an additional check.

2.3 Other spending, mobility, and price data

- **Vehicle purchases.** Monthly vehicle title registration data at the ZIP-code level come from Cross-Sell for California, Texas, Florida, Illinois, and Ohio, plus public data for Washington.
- **Vehicles per household.** The American Community Survey provides census-tract-level data on vehicles per household, which the paper uses as a broader national check.
- **Consumer spending.** Mastercard Center for Inclusive Growth data rank each census tract's annual spending per capita in the national distribution and release the tract's percentile rank.
- **Consumer visits.** SafeGraph data provide weekly visits from residents of a census tract to many types of retail and service locations.
- **Inflation.** The BLS releases bi-monthly regional CPI data by CBSA. Coverage is limited to 23 CBSAs, but it allows the paper to test whether fraud-heavy metros experienced more inflation.

2.4 Controls and competing explanations for house-price growth

- **Demographic controls.** Income, poverty, population density, racial composition, educational attainment, housing units, and vacancy rates are mainly from the American Community Survey.
- **Competing COVID housing channels.** The paper constructs proxies used in the recent literature: remote work before COVID, teleworkability, migration during 2020–21, distance to central business districts, prior house-price growth, and land unavailability.
- **Policy and macro controls.** The analysis also checks robustness to broader PPP take-up, mortgage forbearance, and stimulus checks.

2.5 Background magnitudes and why the mechanism is plausible

- **Estimated PPP fraud.** Griffin et al. (2023) estimate roughly $117.3 billion of PPP fraud, or 14.8% of the program.
- **Other programs.** The SBA OIG identified over $136 billion of potentially fraudulent COVID EIDL loans; a labor-department audit found that about 20% of Pandemic Unemployment Assistance funds in four large states were fraudulent.
- **Overall scale.** The paper argues that total pandemic-relief fraud could plausibly exceed $900 billion, and even a much lower total would still be economically huge.
- **Local size.** A one-standard-deviation increase in flagged PPP loans per capita corresponds to roughly $5 million more suspicious PPP proceeds in an average ZIP code.
- **Why housing can move.** The paper compares that $5 million local shock to roughly $18 million in aggregate housing down payments in a typical ZIP code over 2020–21. That back-of-the-envelope comparison makes it plausible that even partial spending of fraudulent funds into housing can move prices.

3 Models and empirical specifications (all core equations + interpretation)

3.1 House-purchase difference-in-differences

The paper's first individual-level design compares house purchases by recipients of flagged and non-flagged PPP loans:

$$\text{Purchase}_{i,\tau} = \beta\,\mathbf{1}\{\text{Post}_\tau\}\mathbf{1}\{\text{Flagged}_i\} + \gamma\,\mathbf{1}\{\text{Post}_\tau\} + \text{LoanFE}_i + \text{MonthOfYearFE}_\tau + \varepsilon_{i,\tau}. \quad (1)$$

- **Interpretation.** The dependent variable equals one if borrower i buys a house in month τ. The post indicator turns on once the borrower's PPP loan is approved.
- **Key coefficient.** β is the annualized difference in purchase probability for flagged-loan recipients relative to non-flagged-loan recipients.
- **Main result.** The estimated effect is about 0.86 ppt per year, from a baseline annual house-purchase rate of 5.01%. That is a 17% increase.
- **Economic reading.** Fraudulent PPP funds behave like a windfall: recipients are more likely to convert that windfall into housing demand.

3.2 Moving-propensity event study

To compare suspicious PPP borrowers with similar non-recipients living in the same census block group, the paper estimates

$$\begin{aligned}\text{Move}_{i,\tau} = &\sum_{\tau\neq -5} \beta_\tau\,\mathbf{1}\{\text{Period}=\tau\}\mathbf{1}\{\text{SuspiciousPPP}_i\} \\ &+ \text{LoanBySuspiciousFE}_i + \text{LoanByRelativeQuarterFE}_\tau + \varepsilon_{i,\tau}.\end{aligned} \quad (2)$$

- **Interpretation.** The coefficients β_τ trace quarterly moving rates before and after the suspicious loan relative to the omitted period.
- **Pre-trend logic.** If suspicious borrowers were already on a different housing path, the pre-period coefficients should show strong systematic differences.
- **Main result.** Pre-trends are limited; after loan receipt, moving propensity rises sharply. The average post-period effect is about 0.579 ppt per quarter, roughly a 22% increase relative to comparable non-recipients.

3.3 Baseline ZIP-level house-price regression

The core spatial regression is

$$\text{HPGrowth}_i^{2020\text{-}21} = \beta\,\text{FraudPerCapita}_i + \text{CountyFE}_i + \Gamma' X_i + \varepsilon_i. \quad (3)$$

- **Dependent variable.** Cumulative house-price growth in ZIP code i from January 2020 to December 2021.

- **Regressor of interest.** Usually *Flagged Per Capita*, though the paper also uses the alternative fraud proxies and a composite fraud measure.
- **Controls.** Previous 2018–19 house-price growth, PPP loans per capita, population density, vacancy, housing units, and income, usually with population-weighted WLS and county fixed effects.
- **Interpretation of β.** A positive coefficient means fraud-heavy ZIP codes had faster house-price growth than lower-fraud ZIP codes inside the same county.
- **Main result.** In the baseline specification, $\beta = 0.0211$, so a one-standard-deviation increase in flagged loans per capita predicts 2.11 ppt more house-price growth.

3.4 Five-mile neighborhood version

Because buyers can shop outside their home ZIP code, the paper also estimates analogous regressions using local neighborhood averages of fraud:

$$\text{FraudNeighborhood}_i = \sum_{j \in \mathcal{N}(i, 5\text{ miles})} \omega_{ij}\, \text{FraudPerCapita}_j, \tag{4}$$

where ω_{ij} is either an equal weight or a distance weight proportional to $1/(1 + \text{distance}_{ij})$.

- **Interpretation.** This captures the idea that local housing-market pressure is broader than a ZIP boundary.
- **Main result.** The coefficients remain very similar to baseline: about 2.08 to 3.16 ppt depending on the fraud proxy and weighting scheme.

3.5 Instrumental-variables strategy using social connections

The instrument is social proximity to fraud in distant ZIP codes:

$$\text{SocialProximityToFraud}_i = \frac{\sum_{j:\, CBSA(i) \neq CBSA(j)} \text{SCI}_{ij}\, \text{FlaggedPerCapita}_j}{\sum_{j:\, CBSA(i) \neq CBSA(j)} \text{SCI}_{ij}}. \tag{5}$$

The first and second stages are

$$\text{FlaggedPerCapita}_i = \gamma\, \text{SocialProximityToFraud}_i + \text{CountyFE}_i + \Pi' X_i + u_i, \tag{6}$$

$$\text{HPGrowth}_i^{2020\text{-}21} = \beta\, \widehat{\text{FlaggedPerCapita}}_i + \text{CountyFE}_i + \Gamma' X_i + \nu_i. \tag{7}$$

- **Interpretation.** The paper uses social ties to distant fraud clusters as a source of variation in local fraud, motivated by earlier evidence that pandemic fraud spread through social networks.
- **Why distance matters.** Restricting the instrument to ZIP codes outside the same CBSA— and even to ZIP codes more than 500 miles away — reduces concerns that local omitted factors or migration explain the result.
- **Main result.** IV coefficients are around 3.36 to 3.53 ppt across the main distance definitions, larger than the baseline WLS coefficient.

- **Lagged IV refinement.** When the paper instruments 2021 fraud using social proximity to 2020 fraud and controls for the ZIP code's own 2020 fraud and 2020 price growth, the estimate remains large at 2.87 ppt.

3.6 Demand-metric event studies in housing markets

For listing speed, purchases above list price, views per property, and inventory, the paper estimates monthly event-study regressions of the form

$$\begin{aligned}\text{Metric}_{i,t} = &\sum_{t\neq \text{Feb 2020}} \beta_t \,\mathbf{1}\{\text{Month} = t\}\text{FlaggedPerCapita}_i \\ &+ \text{ZIPFE}_i + \text{CountyMonthFE}_{c,t} + \text{Month}_t \times X_i + \varepsilon_{i,t}.\end{aligned} \tag{8}$$

- **Interpretation.** These are reduced-form demand proxies. If fraud created a demand shock, it should first show up in hotter market conditions before reversing later.
- **Main pattern.** Fraud-heavy areas become systematically tighter housing markets in mid-2020 and 2021, then looser housing markets in 2022–23.

3.7 Vehicle-purchase event study

The ZIP-month vehicle-registration regression is

$$\begin{aligned}\log(\text{VehicleRegistrations}_{i,t}) = &\sum_{t\neq \text{Feb 2020}} \beta_t \,\mathbf{1}\{\text{Month} = t\}\text{FlaggedPerCapita}_i \\ &+ \text{ZIPFE}_i + \text{CountyMonthFE}_{c,t} + \varepsilon_{i,t}.\end{aligned} \tag{9}$$

- **Interpretation.** The coefficients trace how vehicle-purchase activity differs over time between high-fraud and low-fraud ZIP codes.
- **Main result.** Fraud-heavy ZIP codes experience about 2.08% higher vehicle title registrations over March 2020 to December 2021 in the event-study averages, and about 2.29% to 3.10% in the collapsed post-period regressions depending on controls.

3.8 Consumer-spending panel

At the census-tract level, the paper estimates

$$\text{SpendingPercentile}_{i,t} = \sum_{t\neq 2019} \beta_t \,\mathbf{1}\{\text{Year} = t\}\text{FlaggedPerCapita}_i + \text{TractFE}_i + \text{CountyYearFE}_{c,t} + \varepsilon_{i,t}. \tag{10}$$

- **Interpretation.** The dependent variable is the national percentile rank of annual consumer spending per capita in the tract.
- **Main result.** A one-standard-deviation increase in fraud raises spending percentile by about 0.88 in 2020 and 0.93 in 2021 relative to 2019, with normalization by 2022.

- **Collapsed-post result.** In the tract-level post regression, the effect is 0.874 percentile points baseline, 1.039 controlling for total PPP take-up, and 0.541 after adding interactions with demographic characteristics.

3.9 Weekly consumer-mobility regression

For SafeGraph visits to different categories of stores and service locations, the paper estimates an analogous weekly specification:

$$\begin{aligned}\Big[\text{IHS}(\text{Visits}_{i,t,\ell}) - \text{IHS}(\text{Devices}_{i,t})\Big] = & \sum_{t \neq \text{week of Feb 17, 2020}} \beta_t \, \mathbf{1}\{\text{Week} = t\}\text{FlaggedPerCapita}_i \\ & + \text{TractFE}_i + \text{CountyWeekFE}_{c,t} \\ & + \delta \, \text{Post}_t \times \text{LoansPerCapita}_i + \varepsilon_{i,t,\ell}.\end{aligned} \tag{11}$$

- **Interpretation.** This traces fraud-associated changes in visits to auto dealers, furniture stores, restaurants, finance, and many other categories.
- **Main result.** Visits rise broadly across nearly all categories after the onset of pandemic relief, consistent with a wide spending response rather than a housing-only channel.

3.10 Regional inflation regression

Finally, for CBSA-level inflation, the paper estimates

$$\begin{aligned}\text{Inflation}^{\text{12-month}}_{i,t} = & \sum_{t \neq \text{Jan/Feb 2020}} \beta_t \, \mathbf{1}\{\text{BiMonth} = t\}\text{FlaggedPerCapita}_i \\ & + \text{CBSAFE}_i + \text{BiMonthFE}_t + \varepsilon_{i,t}.\end{aligned} \tag{12}$$

- **Interpretation.** The coefficients measure whether metro areas with more fraud experienced higher 12-month inflation after COVID.
- **Main result.** A one-standard-deviation increase in fraud is associated with 0.429 ppt higher overall inflation and 0.707 ppt higher housing inflation during the post-March/April 2020 period.
- **Economic intuition.** Since housing is local and immovable, the strongest inflationary effect should show up there first, which is exactly what the data suggest.

4 Identification and credibility

4.1 Core identification logic

- **Why fraud is informative.** The paper's basic idea is not merely that spending rose during COVID. The key distinction is between legitimate transfers, which offset lost income or revenue, and fraudulent transfers, which create excess purchasing power.
- **Why within-county comparisons matter.** County fixed effects absorb local COVID policies, broad local labor-market conditions, and other county-wide shocks. The identification therefore comes from differences in fraud intensity across ZIP codes inside the same county.

- **Why the geography helps.** Fraud is highly concentrated and varies enormously even within counties. That gives the paper rich cross-sectional variation that is much sharper than state-level or metro-level comparisons.

4.2 Main threats and the paper's responses

- **Threat 1: pre-existing housing momentum.** Perhaps high-fraud ZIP codes were already on a stronger house-price path.
- **Response.** The paper controls for 2018–19 house-price growth non-parametrically, builds synthetic controls matched on 2018–19 monthly paths, and shows that treated and control ZIP codes only diverge in summer 2020, not before.
- **Threat 2: omitted local characteristics.** Perhaps fraud is just correlated with local demographics or local COVID-era economic strength.
- **Response.** The paper shows very strong common support across fraud terciles, includes rich demographic controls, runs many sample splits, and uses Oster-style coefficient-stability calculations showing that omitted variables would need implausibly large influence to eliminate the result.
- **Threat 3: reverse causality.** Perhaps house-price growth itself encouraged more fraud.
- **Response.** The lagged IV design instruments 2021 fraud with social proximity to 2020 fraud and still produces a large estimate.
- **Threat 4: migration or spatial spillovers.** Buyers may purchase homes near, but not inside, their home ZIP code.
- **Response.** The five-mile-radius fraud measures deliver similar effects. This strengthens the interpretation that the housing-market shock is genuinely local rather than an artifact of arbitrary ZIP boundaries.

4.3 IV strategy and exclusion restriction

- **Instrument relevance.** Earlier work by the same authors shows that social proximity is the strongest predictor of the spread of pandemic fraud. That gives a natural first-stage for local fraud intensity.
- **Exclusion logic.** Social ties to distant fraud-heavy ZIP codes should affect local house prices mainly through their effect on local fraud take-up. Restricting the instrument to more distant ZIP codes weakens alternative channels like local demand spillovers, migration, or regional shocks.
- **Overidentification test.** When the authors use multiple non-overlapping distance bands, the Hansen J-test does not reject. That does not prove the exclusion restriction, but it is helpful evidence that the instrument is not obviously driven by one particular distance band's direct effect.

4.4 Mechanism evidence

- **Recipient-level evidence.** Fraudulent recipients buy more houses and move more often.

- **Market-tightness evidence.** High-fraud places see more fast sales, more bidding above list, more views, and lower inventory in 2020–21.
- **Supply-elasticity evidence.** The house-price effect is more than 30% larger in less elastic housing markets, exactly as a demand-shock story predicts.
- **Reversal evidence.** When the wave of fraudulent spending fades, demand metrics reverse and house-price growth in high-fraud areas underperforms. That reversal is difficult to reconcile with a purely permanent-fundamentals story.

4.5 Main limitation

- **Not a randomized design.** The paper is careful, but fraud is not randomly assigned, and the interpretation remains quasi-causal rather than experimental.
- **Broader-fraud bundle.** Because PPP fraud is geographically correlated with EIDL and unemployment-insurance fraud, the measured PPP effect likely partly captures the broader local shock from pandemic-relief fraud, not only PPP fraud narrowly defined.
- **Why the evidence is still compelling.** The same story appears in three layers of evidence — recipient behavior, ZIP-level house prices, and other spending/inflation outcomes — and the timing, IV results, and reversal patterns all line up with one coherent mechanism.

5 Tables and figures

5.1 Figure 1: Effect on housing purchases and moving propensity

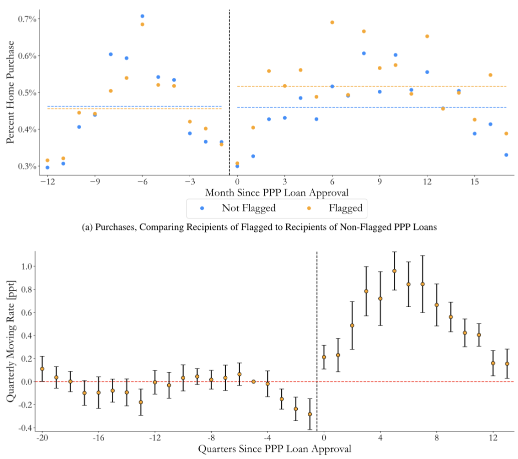
Read:

- **Panel A.** Before loan receipt, flagged and non-flagged recipients have similar house-purchase rates. After receiving the loan, flagged recipients show a clear step-up in purchasing.
- **Panel B.** Suspicious-loan recipients do not look very different from matched non-recipients far before the loan, but their moving rate jumps after loan receipt and stays elevated for multiple quarters.
- **Takeaway.** The figure is the first link in the paper's mechanism: fraudulent recipients appear to convert relief-fraud proceeds into housing demand.

5.2 Table 1: Effect on housing purchases and moving propensity

Read:

- **Purchases.** The difference-in-differences estimate is about 0.0086 annualized, or 0.86 ppt, implying a 17% increase in house purchases for flagged-loan recipients relative to non-flagged recipients.
- **Moving.** The post-period moving coefficient is 0.00579 per quarter when suspicious loans are defined with one primary flag, and it becomes larger under stricter flag definitions.
- **Why it matters.** This table makes the micro-to-macro bridge possible: local fraud can move local prices only if recipients actually spend on housing, and the table shows they do.



(a) Purchases, Comparing Recipients of Flagged to Recipients of Non-Flagged PPP Loans

(b) Moving, Comparing Recipients of Flagged PPP Loans to Non-Recipients

Fig. 1. Effect on Housing Purchases and Moving Propensity. This figure shows the relationship between individuals receiving a flagged PPP loan and their likelihood of housing purchase (Panel A) and moving (Panel B). In Panel A, data from PropertyRadar for a sample of 250,000 loans is used to determine home purchases from one year before the individual received their PPP loan to 18 months after. The sample of non-flagged loans is reweighted to match the distribution of the timing of PPP loan approval in the sample of flagged loans. The horizontal lines are the average monthly likelihood of an individual in each group buying a house during the pre-/post-period. In Panel B, address history data from Verisk is used to track the moving propensity of recipients of PPP loans flagged by at least one primary flag and a comparison set of individuals who likely did not receive a PPP loan and were living in the same census block group as the recipient when the PPP loan was received. For each PPP recipient and their corresponding comparison set, we track moving rates from twenty quarters before the PPP loan was received to thirteen quarters afterward. To ensure correct weighting, a simple average of movement rates for each quarter across all addresses in each loan's comparison set is taken. The quarter that is five periods before the PPP loan was received is the omitted period, and the regression includes loan × 1(Suspicious PPP) and loan × relative quarter fixed effects. The error bars represent 95% confidence intervals based on standard errors that are double clustered by county and week of PPP approval. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Effect on housing purchases and moving propensity.

Panel A: Purchases, comparing recipients of flagged to recipients of non-flagged PPP loans				
	1(Housing purchase during month) × 12			
	PropertyRadar		LexisNexis	
	(1)	(2)	(3)	(4)
1(Post) × 1(Flagged)	0.00863***	0.00865***	0.0109***	0.00790***
	(5.22)	(5.59)	(4.44)	(4.87)
1(Post)	0.00686**		−0.00517	
	(2.16)		(−1.24)	
Loan FE	Yes	Yes	Yes	Yes
Month of Year FE	Yes	Yes	Yes	Yes
1(Post) × ln(Loan amount)	No	Yes	No	Yes
1(Post) × County FE	No	Yes	No	Yes
1(Post) × Business type FE	No	Yes	No	Yes
1(Post) × Week approved FE	No	Yes	No	Yes
Observations	19,500,000	19,500,000	9,600,000	9,600,000
R^2	0.0279	0.0279	0.0202	0.0206

Panel B: Moving, Comparing recipients of flagged PPP loans to non-recipients			
	1(Moved during quarter)		
	Primary (1)	Primary + Additional (2)	Two Primary (3)
1(Post) × 1(Suspicious PPP)	0.00579***	0.00644***	0.00848***
	(9.43)	(10.95)	(8.21)
Loan × 1(Suspicious PPP) FE	Yes	Yes	Yes
Loan × Relative Quarter FE	Yes	Yes	Yes
Observations	26,964,652	19,272,016	2,669,000
R^2	0.534	0.534	0.534

This table examines whether individuals purchased homes (Panel A) and moved (Panel B) after receiving a flagged PPP loan. In Panel A, data on home purchases for a sample of 250,000 loans from PropertyRadar is used in columns (1) and (2) and for a sample of 150,000 loans from LexisNexis in columns (3) and (4). For each individual, we include monthly observations for the five years before they received their PPP loan to 18 (4) months after for PropertyRadar (LexisNexis). 1(*Flagged*) takes a value of 1 if the PPP loan is flagged by at least one primary flag. In Panel B, address history data from Verisk is used to track the moving propensity of recipients of flagged PPP loans and a comparison set of individuals who likely did not receive a PPP loan and were living in the same census block group as the recipient when the PPP loan was received. For each PPP recipient and their corresponding comparison set, we include quarterly observations from twenty quarters before the PPP loan was received to thirteen quarters afterward. To ensure correct weighting, a simple average of movement rates for each quarter across all addresses in each loan's comparison set is taken. Column (1) is based on loans flagged by at least one primary flag, column (2) on loans flagged by at least one primary and an additional (either primary or secondary) flag, and column (3) is based on loans flagged by at least two primary flags. Fixed effects and controls are indicated at the bottom of each column. t-statistics based on robust standard errors that are double clustered by PPP loan and month in Panel A and by county and week of PPP approval in Panel B are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

5.3 Table 2: Effect on house-price growth in 2020–21

Read:

- **Column 1.** Without controls other than county fixed effects, a one-standard-deviation increase in flagged loans per capita predicts 1.79 ppt more house-price growth.
- **Column 2.** In the baseline controlled specification, the effect rises to 2.11 ppt.
- **Columns 3–5.** Alternative fraud measures are similar or stronger. The broad composite measure gives 2.68 ppt.
- **Takeaway.** House prices rose materially faster in fraud-heavy ZIP codes, and the result is not tied to one narrow fraud proxy.

5.4 Table 3: Five-mile-radius fraud measures

Read:

- Fraud in neighboring ZIP codes within five miles has nearly the same predictive power for house prices as the focal ZIP's own fraud rate.
- Simple averages and distance-weighted averages both work.
- This is important because housing markets are local neighborhoods, not strict ZIP-code islands.

5.5 Figure 2: Effect on house-price growth

Read:

Table 2
Effect on house price growth.

	House price growth from January 1, 2020 to December 31, 2021				
	(1)	(2)	(3)	(4)	(5)
Flagged Per Capita	0.0179*** (10.38)	0.0211*** (9.84)			
High Loan-to-Est. Per Capita			0.0222*** (11.97)		
High Similarity Per Capita				0.0224*** (11.06)	
Flagged Composite Per Capita					0.0268*** (13.34)
County FE	Yes	Yes	Yes	Yes	Yes
Past HP Growth	No	Yes	Yes	Yes	Yes
Loans Per Capita	No	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Observations	18,761	18,761	18,761	18,761	18,761
Num. Counties	2215	2215	2215	2215	2215
R^2	0.781	0.828	0.824	0.828	0.827
Mean of Dep. Var.	0.259	0.259	0.259	0.259	0.259

This table examines the relationship between house price growth and various measures of suspicious PPP lending. *Flagged Per Capita* is a ratio of the number of flagged PPP loans in the ZIP code to the ZIP code's population. *High Loan-to-Est. Per Capita* is based on the number of PPP loans in county-industry pairs where there are more than two times as many PPP loans as establishments per the US Census CBP. *High Similarity Per Capita* is based on the number of PPP loans in county-lender pairs with high levels of similarity in terms of loan amount, jobs reported, and industry. *Flagged Composite Per Capita* is based on the number of loans that are either flagged, in industry-counties where there are more than two times as many PPP loans as establishments, or in lender-counties with high levels of similarity. See Griffin et al. (2023) for additional details about these measures. *Past HP Growth* and *Loans Per Capita* control for house price growth in 2018–19 and PPP lending intensity, respectively, using percentile fixed effects. The controls included are log population density, vacancy rate, log housing units, and log average household income. All independent variables are standardized to have a mean of 0 and a standard deviation of 1. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the population of the ZIP code in 2019. Fixed effects are as indicated at the bottom of each column. t-statistics based on robust standard errors that are clustered by county are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 3
Effect on house price growth, suspicious lending in five-mile radius.

	House price growth from January 1, 2020 to December 31, 2021			
	Simple average		Distance weighted average	
	(1)	(2)	(3)	(4)
Flagged Per Capita	0.0208*** (10.74)		0.0243*** (12.08)	
Flagged Composite Per Capita		0.0262*** (8.63)		0.0316*** (11.35)
County FE	Yes	Yes	Yes	Yes
Past HP Growth	Yes	Yes	Yes	Yes
Loans Per Capita	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	18,761	18,761	18,761	18,761
Num. Counties	2215	2215	2215	2215
R^2	0.824	0.824	0.826	0.826
Mean of Dep. Var.	0.259	0.259	0.259	0.259

This table examines the relationship between house price growth and average suspicious lending rates in a five-mile radius around a ZIP code. In columns (1) and (2), simple averages of suspicious lending rates among all ZIP codes in a five-mile radius of the focal ZIP code are used. In columns (3) and (4), distance-weighted averages (specifically, weighted by 1/(1 + distance)) of suspicious lending rates among all ZIP codes in a five-mile radius of the focal ZIP code are used. *Past HP Growth* and *Loans Per Capita* control for house price growth in 2018–19 and PPP lending intensity, respectively, using percentile fixed effects. The controls included are log population density, vacancy rate, log housing units, and log average household income. All independent variables are standardized to have a mean of 0 and a standard deviation of 1. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the population of the ZIP code in 2019. Fixed effects are as indicated at the bottom of each column. t-statistics based on robust standard errors that are clustered by county are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

- **Panel A.** The binscatter shows a close-to-linear positive relation between fraud per capita and 2020–21 house-price growth. The difference between the lowest and highest fraud deciles is 5.8 ppt.
- **Panel B.** The fraud coefficient turns positive quickly, becomes significant in April 2020, peaks in June 2022, and then shrinks through 2023.
- **Panel C.** The effect is similar across many demographic splits, which weakens the concern that the result is driven by one narrow type of neighborhood.
- **Takeaway.** The figure shows both timing and breadth: the effect starts when fraud could plausibly matter, and it is not confined to a single demographic subgroup.

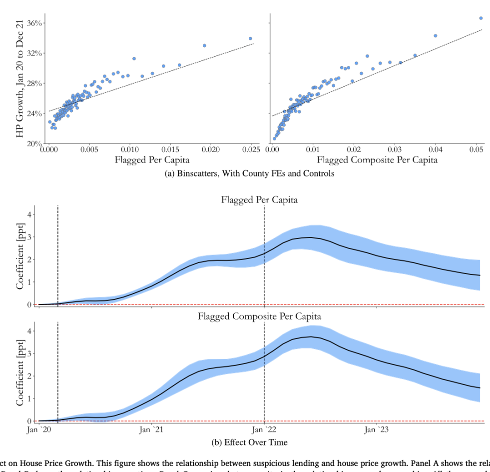


Fig. 2. Effect on House Price Growth. This figure shows the relationship between suspicious lending and house price growth. Panel A shows the relationship using binscatters. Panel B shows the relationship over time. Panel C examines heterogeneity in the relationship across demographics. All three panels are estimated using ZIP code-level data, include county fixed effects, and control for house price growth in 2018 to 2019 and loans per capita using percentile fixed effects. Further, they control for log population density, vacancy rate, log housing units, and log average household income. The left subpanel of Panel A and the top subpanels of Panels B and C are based on *Flagged Per Capita*. The right subpanel of Panel A and the bottom subpanels of Panels B and C are based on *Flagged Composite Per Capita*. *Flagged Per Capita* is a ratio of the number of flagged PPP loans in the ZIP code to the ZIP code's population. *Flagged Composite Per Capita* is based on the number of loans that are either flagged, in county-industry pairs where there are more than two times as many PPP loans as establishments, or in county-lender pairs with high levels of similarity. See Griffin et al. (2023) for additional details about these measures. In Panels B and C, the measures of suspicious lending are standardized, so the coefficients represent the house price effect of a one standard deviation change in suspicious lending. The splits in Panel C are based on the median value of the demographic. To have a nationally representative estimate, all three panels use weighted least squares (WLS) regressions with the weight being the population of the ZIP code in 2019. The error bars in Panels B and C represent 95% confidence intervals based on standard errors clustered by county.

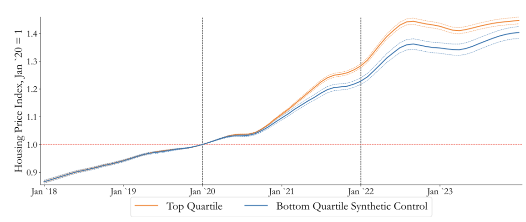


Fig. 3. Effect on House Price Growth, Synthetic Control. This figure shows the effect of suspicious lending on house price growth. A synthetic control method is used to create controls for each ZIP code in the top quartile of *Flagged Per Capita* using all ZIP codes in the same county that are in the bottom quartile of *Flagged Per Capita*. ZIP codes are split into quartiles within each county. Counties where the difference between the 75th and 25th percentiles of Flagged Per Capita within the county is at least half as large as the standard deviation across the entire nation are included. The dashed lines are 95% confidence intervals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.6 Figure 3: Synthetic-control evidence on house prices

Read:

- The synthetic-control series for low-fraud ZIP codes is almost identical to the treated high-fraud series through 2018, 2019, and early 2020.
- The divergence begins only in summer 2020, exactly when fraudulent funds could reasonably begin to affect the housing market.
- By the end of 2021, top-quartile-fraud ZIP codes reach about 28.3% cumulative house-price growth versus 22.8% for their synthetic controls.
- This is among the most persuasive pieces of evidence in the paper because it directly addresses pre-trend concerns.

5.7 Table 4: Balance across fraud terciles and social-proximity terciles

Read:

- High-fraud ZIP codes differ somewhat from low-fraud ZIP codes, but most differences are modest relative to the cross-sectional standard deviation of the variable.
- For fraud terciles, even the largest standardized mean difference is only about 0.36 standard deviations.
- For the social-proximity instrument, differences are somewhat larger, but the distributions still overlap heavily.
- The table supports the paper's claim of broad common support and helps motivate both the control strategy and the IV design.

Table 4
Balance tables by within-county terciles of Flagged Per Capita and social proximity to suspicious lending.

Panel A: Within-county terciles of Flagged Per Capita

	Low fraud		Medium fraud		High fraud		High vs. Low		
	Mean	Median	Mean	Median	Mean	Median	Diff.	SE	Diff./SD
Log(Income)	11.117	11.042	11.107	11.022	11.076	10.979	−0.041***	(0.015)	−0.092
Pct. Poverty	0.112	0.095	0.125	0.109	0.143	0.125	0.031***	(0.002)	0.364
Log(Population density)	5.755	5.350	5.755	5.436	5.559	5.282	−0.197***	(0.045)	−0.097
Pct. Non-White	0.151	0.095	0.183	0.119	0.223	0.136	0.071***	(0.007)	0.346
Educational attainment	0.279	0.236	0.286	0.238	0.282	0.229	0.003	(0.005)	0.020
Pre-Pandemic unemployment	0.049	0.044	0.050	0.045	0.054	0.048	0.005***	(0.001)	0.172
Log(Distance to CBD)	2.803	2.872	2.630	2.719	2.478	2.666	−0.325***	(0.021)	−0.334
Land unavailability	0.256	0.250	0.254	0.238	0.258	0.237	0.002	(0.004)	0.011
Remote work 2015–19	0.052	0.046	0.051	0.045	0.053	0.044	0.001	(0.001)	0.026
Teleworkable	0.360	0.353	0.358	0.349	0.352	0.344	−0.008***	(0.001)	−0.173
Net migration 2020–21	−0.001	−0.002	−0.006	−0.006	−0.010	−0.011	−0.009***	(0.002)	−0.107
House price growth 2018–19	0.109	0.104	0.110	0.106	0.116	0.109	0.007***	(0.002)	0.098

Panel B: Within-county terciles of social proximity to suspicious lending

	Low SP		Medium SP		High SP		High vs. Low		
	Mean	Median	Mean	Median	Mean	Median	Diff.	SE	Diff./SD
Log(Income)	11.188	11.097	11.144	11.042	11.006	10.936	−0.182***	(0.014)	−0.417
Pct. Poverty	0.100	0.085	0.119	0.103	0.154	0.139	0.054***	(0.002)	0.627
Log(Population density)	5.449	5.027	5.693	5.383	5.783	5.570	0.334***	(0.048)	0.166
Pct. Non-White	0.125	0.076	0.167	0.111	0.250	0.168	0.126***	(0.006)	0.590
Educational attainment	0.291	0.246	0.296	0.243	0.268	0.222	−0.022***	(0.005)	−0.139
Pre-Pandemic unemployment	0.047	0.041	0.048	0.044	0.057	0.051	0.011***	(0.001)	0.332
Log(Distance to CBD)	2.940	2.977	2.641	2.723	2.372	2.518	−0.568***	(0.026)	−0.589
Land unavailability	0.256	0.248	0.255	0.239	0.257	0.237	0.002	(0.004)	0.010
Remote work 2015–19	0.058	0.051	0.053	0.046	0.048	0.041	−0.010***	(0.001)	−0.259
Teleworkable	0.363	0.355	0.360	0.350	0.349	0.342	−0.014***	(0.001)	−0.326
Net migration 2020–21	0.010	0.003	−0.003	−0.004	−0.018	−0.017	−0.028***	(0.002)	−0.355
House price growth 2018–19	0.104	0.100	0.107	0.104	0.122	0.114	0.018***	(0.002)	0.252

This table examines differences in various ZIP code-level socioeconomic variables and factors the literature has found to be associated with house price growth during 2020–21 by the level of suspicious lending in the ZIP code (Panel A) and the ZIP code's social proximity to suspicious lending (Panel B). ZIP codes in each county are split into terciles based on their ratio of flagged loans to population (Panel A) and on their social proximity to suspicious lending (Panel B). Mean and median values of each variable among ZIP codes in each tercile are provided. In addition, the difference in the mean value of the variable between the low- and high-fraud terciles, the standard error for the difference, and the standardized mean difference (difference divided by the standard deviation) are provided. Robust standard errors are clustered by county and are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 5
Effect on house price growth, IV.

	House price growth in 2020–21					2021 only
	Outside CBSA (1)	≥100 Mi (2)	≥250 Mi (3)	≥500 Mi (4)	Concentric rings (5)	2020 Outside CBSA (6)
Flagged Per Capita	0.0347***	0.0353***	0.0336***	0.0340***	0.0367***	
	(6.35)	(6.51)	(6.17)	(5.43)	(6.41)	
2021 Flagged Per Capita						0.0287***
						(2.76)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Past HP Growth	Yes	Yes	Yes	Yes	Yes	Yes
Loans Per Capita	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
2020 Flagged Per Capita	No	No	No	No	No	Yes
2020 House price growth	No	No	No	No	No	Yes
Observations	16,869	18,761	18,761	18,761	18,725	16,868
Num. Counties	1789	2215	2215	2215	2213	1789
Mean of Dep. Var.	0.257	0.259	0.259	0.259	0.259	0.148
First stage F-stat	34.33	38.04	34.27	28.48	14.38	14.81
Hansen's J-stat					2.465	
(p-value)					(0.292)	

This table examines the relationship between house price growth and suspicious PPP lending using instrumented variables based on social connectedness between ZIP codes. Column (1) is based on social connectedness between each ZIP code and ZIP codes that are outside the given ZIP code's CBSA. Columns (2), (3), and (4) are based on social connectedness between each ZIP code and ZIP codes that are at least 100, 250, and 500 miles away, respectively. Column (5) includes three instruments based on social connections to ZIP codes in non-overlapping rings (between 100 and 250 miles, between 250 and 500 miles, and over 500 miles) at the same time. The J-stat and p-value for an overidentification test are provided at the bottom of column (5). Column (6) is based on fraud rates in 2020 and social connectedness between each ZIP code and ZIP codes that are outside the given ZIP code's CBSA. Columns (1) to (5) use house price growth during the entire 2020–21 period as the dependent variable, whereas column (6) uses house price growth during only 2021 as the dependent variable. *Past HP Growth* and *Loans Per Capita* control for house price growth in 2018–19 and PPP lending intensity, respectively, using percentile fixed effects. The controls included are log population density, vacancy rate, log housing units, log average household income, and the share of Facebook friends within 50 and 150 miles. *2020 Flagged Per Capita* and *2020 House Price Growth* are the fraud rate and house price growth for each ZIP code during 2020. All independent variables are standardized to have a mean of 0 and a standard deviation of 1. To have a nationally representative estimate, we use weighted least squares (WLS) regressions in both stages, with the weight being the population of the ZIP code in 2019. Fixed effects are as indicated at the bottom of each column. t-statistics based on robust standard errors that are clustered by county are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

5.8 Table 5: IV estimates of the house-price effect

Read:

- Using social proximity to fraud outside the local CBSA, the IV estimate is 3.47 ppt.

- Restricting the instrument to ZIP codes at least 100, 250, or 500 miles away leaves the estimate essentially unchanged: about 3.4 to 3.5 ppt.
- The 2021-only lagged-IV exercise still gives 2.87 ppt.
- This table says the baseline result is not only a descriptive correlation. It survives a demanding strategy based on distant social spillovers in fraud take-up.

5.9 Table 6: Univariate and multivariate horse race

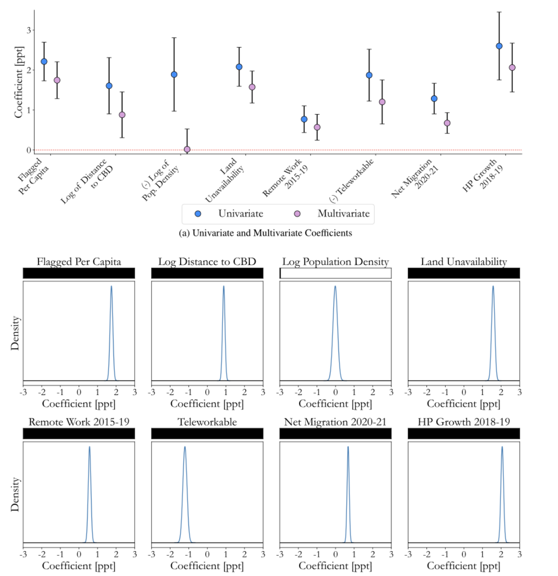
Read:

- In univariate regressions, fraud, land unavailability, prior price growth, teleworkability, migration, and other proposed variables all matter.
- In the multivariate regression, population density loses significance, but flagged fraud remains large and highly significant at 1.75 ppt.
- The other large multivariate predictors are prior house-price growth (2.06 ppt) and land unavailability (1.57 ppt).
- This is the paper's key comparison-to-the-literature table: fraud remains one of the most important explanations even after stacking the other channels against it.

Table 6
Effect on house price growth, univariate and multivariate.

	House price growth from January 1, 2020 to December 31, 2021								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flagged Per Capita	0.0221*** (8.99)								0.0175*** (7.41)
Log of Dist. to CBD		0.0161*** (4.47)							0.00880*** (3.01)
Log of Pop. Density			−0.0189*** (−4.03)						−0.000170 (−0.07)
Land unavailability				0.0208*** (8.36)					0.0157*** (7.72)
Remote Work 2015–19					0.00771*** (4.52)				0.00569*** (3.44)
Teleworkable						−0.0187*** (−5.66)			−0.0120*** (−4.27)
Net migration 2020–21							0.0129*** (6.55)		0.00674*** (5.12)
HP Growth 2018–19								0.0260*** (6.00)	0.0206*** (6.61)
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Past HP Growth	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Loans Per Capita	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observation	12,305	12,305	12,305	12,305	12,305	12,305	12,305	12,305	12,305
Num. Counties	1013	1013	1013	1013	1013	1013	1013	1013	1013
R^2	0.814	0.811	0.807	0.810	0.803	0.806	0.810	0.797	0.828
Mean of Dep, Var.	0.257	0.257	0.257	0.257	0.257	0.257	0.257	0.257	0.257

This table examines the univariate and multivariate relationships between various proposed variables and house price growth. *Past HP Growth* and *Loans Per Capita* control for house price growth in 2018–19 and PPP lending intensity, respectively, using percentile fixed effects. The controls included are vacancy rate, log housing units, and log average household income. All independent variables are standardized to have a mean of 0 and a standard deviation of 1. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the population of the ZIP code in 2019. Only ZIP codes for which all variables can be determined are used. Fixed effects are as indicated at the bottom of each column. t-statistics based on robust standard errors that are clustered by county are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.



(a) Univariate and Multivariate Coefficients

(b) Bayesian Model Averaging

Fig. 4. Effect of other proposed variables on house price growth. This figure shows the effect of each proposed variable on house price growth. Panel A shows univariate and multivariate coefficients using WLS regressions. Panel B shows the posterior coefficient distribution, conditional on inclusion, from multivariate regressions using Bayesian model averaging. The black bars at the top of each distribution plot the posterior inclusion probability. All regressions control for vacancy rate, log housing units, log average household income, and for overall PPP loans per capita and house price growth in 2018–19 using percentile indicator variables to allow for non-linearity, and include county fixed effects. All proposed variables are standardized to have a mean of 0 and a standard deviation of 1. The weighted least squares (WLS) regressions are weighted by the population of the ZIP code in 2019. Only ZIP codes for which all proposed variables can be determined are used in both panels. The error bars in Panel A represent 95% confidence intervals based on standard errors clustered by county. The regressions corresponding to Panel A are shown in Table 6. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.10 Figure 4: Comparing proposed drivers of house-price growth

Read:

- **Panel A.** Visually, flagged fraud, land unavailability, and prior 2018–19 price growth are the dominant predictors in the multivariate specification.

- **Panel B.** Bayesian model averaging assigns posterior inclusion probability 1.000 to flagged fraud, land unavailability, prior price growth, teleworkability, migration, remote work, and distance to CBD; population density gets only 0.009.
- The paper is not claiming fraud is the *only* factor. It is claiming fraud consistently belongs in the small set of top factors, and the figure shows that succinctly.

5.11 Table 7: Variable selection using the Bayesian Information Criterion

Read:

- If the model is allowed to include only one proposed channel, the selected variable is flagged fraud.
- As more variables are allowed, fraud remains selected in every optimal model.
- The optimal specification includes seven of the eight proposed channels, excluding only population density.
- The message is robustness: no sensible model-selection rule drops fraud.

Table 7
Effect on house price growth, variable selection.

	House price growth from January 1, 2020 to December 31, 2021								
	Regression coefficients and t-statistics								PIC Prob.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flagged Per Capita	0.0269*** (11.52)	0.0223*** (9.79)	0.0202*** (8.02)	0.0186*** (7.65)	0.0176*** (7.41)	0.0176*** (7.57)	0.0175*** (7.42)	0.0175*** (7.41)	1.000
HP Growth 2018–19		0.0215*** (5.96)	0.0231*** (6.93)	0.0217*** (6.72)	0.0217*** (6.74)	0.0208*** (6.78)	0.0206*** (6.60)	0.0206*** (6.61)	1.000
Log of Dist. to CBD			0.0142*** (3.94)	0.0127*** (3.97)	0.00990*** (3.20)	0.00903*** (2.95)	0.00886*** (2.89)	0.00880*** (3.01)	1.000
Land unavailability				0.0163*** (8.16)	0.0161*** (7.77)	0.0159*** (7.44)	0.0158*** (7.59)	0.0157*** (7.72)	1.000
Net migration 2020–21					0.00825*** (6.20)	0.00740*** (5.74)	0.00676*** (5.42)	0.00674*** (5.12)	1.000
Teleworkable						−0.0108*** (−3.78)	−0.0120*** (−4.27)	−0.0120*** (−4.27)	1.000
Remote work 2015–19							0.00569*** (3.44)	0.00569*** (3.44)	1.000
Log of Pop. Density								−0.000170 (−0.07)	0.009
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Loans Per Capita	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observation	12,305	12,305	12,305	12,305	12,305	12,305	12,305	12,305	12,305
Num. Counties	1013	1013	1013	1013	1013	1013	1013	1013	1013
R^2	0.798	0.811	0.818	0.823	0.826	0.827	0.828	0.828	
Mean of Dep. Var.	0.257	0.257	0.257	0.257	0.257	0.257	0.257	0.257	0.257

This table shows the results of a variable selection process in which the optimal model, with between one and eight of the potential independent variables, is selected based on the Bayesian Information Criterion. Column (9) reports posterior inclusion probabilities for each variable based on Bayesian model averaging. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the ZIP code's population as of 2019. All independent variables are standardized to have a mean of 0 and a standard deviation of 1. Fixed effects are as indicated at the bottom of each column. t-statistics based on robust standard errors that are clustered by county are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

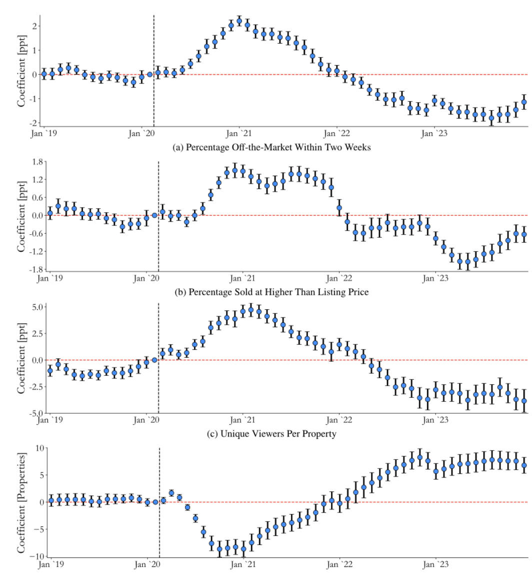


Fig. 5. Effect on other housing market metrics. This figure shows the effect of a one standard deviation change in *Flagged Per Capita* on various housing market metrics. We estimate the following regression and plot β_t: $Metric_{i,t} = \sum_{t \neq Feb2020} \beta_t 1(Month = t) FlaggedPerCapita_i + ZIPCodeFE + CountyMonthFE + Month_t \times Controls_i + \epsilon_{i,t}$ where i is a ZIP code and t is a month. Panel A shows the effect on the percentage of properties off the market within two weeks, Panel B on the percentage of properties sold at higher than the listing price, Panel C on unique viewers per property (relative to a typical property across the US), and Panel D on inventory (the number of active listings). Panels A and B (C and D) are based on metrics from Redfin (Realtor.com). The dependent variable is the difference between the metric during the given month and its value in the same month of 2018 for 2019, and between the metric during the given month and its average value in 2018 and 2019 for 2020 onward. The controls are loans per capita using percentile fixed effects, log population density, vacancy rate, log housing units, and log average household income. The error bars correspond to 95% confidence intervals based on standard errors that are double clustered by ZIP code and month.

5.12 Figure 5: Housing-market tightness and reversal

Read:

- Fraud-heavy ZIP codes begin to look hotter in mid-2020: more homes sell within two weeks, more sell above list, views per property rise, and inventory falls.
- Those differences weaken in 2021 and reverse through 2022–23: fewer quick sales, fewer above-list purchases, fewer views, and more inventory.

- This figure is the cleanest market-mechanism evidence. It shows that high-fraud places were temporarily demand-constrained and later temporarily demand-depressed.

5.13 Table 8: House-price reversal in 2022–23

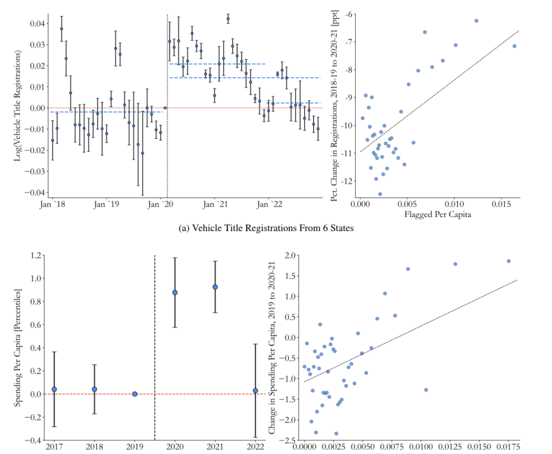
Read:

- The 2020–21 coefficient is positive; the 2022–23 coefficient turns negative.
- For flagged loans per capita, the reversal coefficient is about −0.744 ppt in the baseline controlled specification.
- For the composite fraud measure, the reversal is even larger at −1.09 ppt.
- The table quantifies the paper's out-of-sample prediction that a temporary fraud-driven demand surge should partly unwind.

Table 8
Effect on house price growth, 2022–23.

	House price growth from January 1, 2022 to December 31, 2023				
	(1)	(2)	(3)	(4)	(5)
Flagged Per Capita	−0.00606***	−0.00744***			
	(−4.49)	(−4.40)			
High Loan-to-Est. Per Capita			−0.0124***		
			(−8.13)		
High Similarity Per Capita				−0.00897***	
				(−6.09)	
Flagged Composite Per Capita					−0.0109***
					(−6.02)
County FE	Yes	Yes	Yes	Yes	Yes
Past HP Growth	No	Yes	Yes	Yes	Yes
Loans Per Capita	No	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Observations	18,761	18,761	18,761	18,761	18,761
Num. Counties	2215	2215	2215	2215	2215
R^2	0.810	0.823	0.824	0.823	0.823
Mean of Dep. Var.	0.120	0.120	0.120	0.120	0.120

This table examines the relationship between house price growth in 2022–23 and various measures of suspicious PPP lending. All variables, fixed effects, and controls are as defined in Table 2. All independent variables are standardized to have a mean of 0 and a standard deviation of 1. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the population of the ZIP code in 2019. Fixed effects are as indicated at the bottom of each column. t-statistics based on robust standard errors that are clustered by county are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.



(b) Consumer Spending

Fig. 6. Effect on Vehicle Purchases and Consumer Spending. This figure shows the effect of suspicious lending on vehicle purchases and consumer spending. Panel A uses vehicle title registration data from six states (California, Texas, Florida, Illinois, Ohio, and Washington) at the ZIP code-month level. Panel B uses annual data from 2017 to 2022 at the census tract level from Mastercard's Center for Inclusive Growth. Mastercard ranks each census tract's consumer spending per capita each year in the national distribution and releases the percentile rank of the tract. Data from 2017 to 2022 is used. The left subpanels examine the effects of a one standard deviation change in the number of flagged PPP loans per capita. The left subpanel of Panel A includes ZIP code and month × county fixed effects, and the left subpanel of Panel B includes census tract and year × county fixed effects. The error bars in the left subpanels represent 95% confidence intervals based on standard errors that are double clustered by county and month in Panel A and clustered by county in Panel B. In the left subpanel of Panel A, the horizontal blue lines represent the average coefficient over the period spanned by each line. The right subpanels examine the within-county effects of flagged PPP loans per capita on the percentage change in vehicle registrations between 2018–19 and 2020–21 in Panel A and the percentile change in spending per capita between 2019 and 2020–21 in Panel B. To have a nationally representative estimate, both panels use weighted least squares (WLS) regressions with the weight being the ZIP code's (census tract's) population as of 2019 in Panel A (Panel B). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.14 Figure 6: Vehicle purchases and consumer spending

Read:

- **Panel A.** Vehicle title registrations are elevated in high-fraud ZIP codes during 2020–21, then largely normalize in 2022.
- **Panel B.** Consumer spending percentile ranks are higher in high-fraud tracts in 2020 and 2021, then return close to normal in 2022.
- The right-side binscatters in both panels show that these patterns are not driven by a handful of outliers.
- The figure broadens the paper's message from housing to general local consumption.

5.15 Table 9: Vehicle purchases

Read:

- Over March 2020 to December 2021, flagged fraud predicts 2.29% more vehicle title registrations in the baseline post specification.
- After controlling for total PPP lending and detailed demographic interactions, the effect remains 2.39%.
- Extending the post period through December 2022 yields smaller but still positive coefficients.
- The table shows that fraudulent funds were spent on major durables beyond housing.

Table 9
Effect on vehicle purchases.

	Log(Vehicle title registrations)					
	March 2020 to December 2021			March 2020 to December 2022		
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Flagged Per Capita	0.0229^{***} (5.42)	0.0310^{***} (3.76)	0.0239^{***} (9.94)	0.0163^{***} (4.32)	0.0201^{***} (2.98)	0.0163^{***} (9.56)
Post × Loans Per Capita		-0.0150^{**} (−2.23)	-0.00587^{**} (−2.36)		−0.00708 (−1.24)	−0.00136 (−0.56)
Post × Median income			0.0203^{**} (2.36)			0.0275^{***} (3.40)
Post × Poverty			0.00267 (0.37)			0.00295 (0.46)
Post × Population density			0.0173^{***} (4.12)			0.0146^{***} (3.65)
Post × Pct. Non-White			−0.00786 (−1.57)			−0.00156 (−0.36)
Post × Educational attainment			-0.0285^{**} (−2.47)			-0.0233^{**} (−2.20)
Post × Pre-Pandemic unemployment			0.0104^{**} (2.58)			0.00599 (1.67)
ZIP Code FE	Yes	Yes	Yes	Yes	Yes	Yes
Month × County FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	278,400	278,400	278,400	348,000	348,000	348,000
R^2	0.981	0.981	0.981	0.979	0.979	0.979

This table examines the effect of suspicious lending on vehicle purchases using vehicle title registration data from six states (California, Texas, Florida, Illinois, Ohio, and Washington) at the ZIP code-month level. *Post* is a dummy variable that takes a value of 1 from March 2020 to December 2021 (2022) in columns (1) to (3) (columns (4) to (6)) and a value of 0 from January 2018 to February 2020 in all columns. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the ZIP code's population as of 2019. All independent variables are standardized to have a mean of 0 and a standard deviation of 1. Fixed effects are as indicated at the bottom of each column. *t*-statistics based on robust standard errors that are double clustered by county and month are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 10
Effect on consumer spending.

	Spending Per Capita percentile rank			
	(1)	(2)	(3)	(4)
Post × Flagged Per Capita	0.874^{***} (8.59)	1.039^{***} (10.05)	0.541^{***} (4.08)	0.738^{***} (7.03)
Post × Loans Per Capita		-0.232^{***} (−2.92)	0.0569 (0.52)	−0.103 (−1.50)
Post × Median income			−0.0667 (−0.34)	-0.525^{***} (−3.02)
Post × Poverty			0.0995 (0.72)	0.168 (1.23)
Post × Population density			0.566^{*} (1.92)	0.758^{***} (5.71)
Post × Pct. Non-White			0.523^{***} (3.12)	0.0882 (0.64)
Post × Educational attainment			-1.831^{**} (−1.99)	-1.593^{**} (−1.99)
Post × Pre-Pandemic unemployment			0.185^{*} (1.72)	0.452^{***} (4.39)
Census tract FE	Yes	Yes	Yes	Yes
Year × County FE	Yes	Yes	Yes	No
Year FE	No	No	No	Yes
Observations	307,385	307,385	307,385	307,385
R^2	0.348	0.348	0.348	0.305

This table examines the effect of suspicious lending on consumer spending using annual data at the census tract level from Mastercard's Center for Inclusive Growth. Mastercard ranks each census tract's consumer spending per capita each year in the national distribution and only releases the percentile rank of the tract for each year. Data from 2017 to 2021 is used. *Post* is a dummy variable that takes a value of 1 if the year is 2020 or 2021 and 0 otherwise. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the census tract's population as of 2019. All independent variables are standardized to have a mean of 0 and a standard deviation of 1. Fixed effects are as indicated at the bottom of each column. *t*-statistics based on robust standard errors that are clustered by county are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

5.16 Table 10: Consumer spending

Read:

- The baseline post-period effect is 0.874 percentile points in spending per capita, rising to 1.039 when controlling for overall PPP lending.
- With demographic interactions, the effect remains 0.541 percentile points and strongly significant.
- Replacing county-year fixed effects with year fixed effects makes the coefficient even larger, implying fraud predicts spending both within and across counties.
- This table says the spending response was broad, not merely a housing-market reallocation.

5.17 Figure 7: Consumer visits across location types

Read:

- High-fraud tracts show more visits to automobile dealers, food and beverage stores, furniture and home furnishings stores, clothing stores, jewelry stores, electronics stores, general merchandise stores, sporting-goods stores, gun stores, arts and recreation, accommodation and food services, and finance and insurance.

- The pre-period is mostly flat, while the post-2020 period is broadly and persistently positive.
- This figure suggests that fraud-induced spending was economically wide-ranging and not narrowly confined to one spending category.

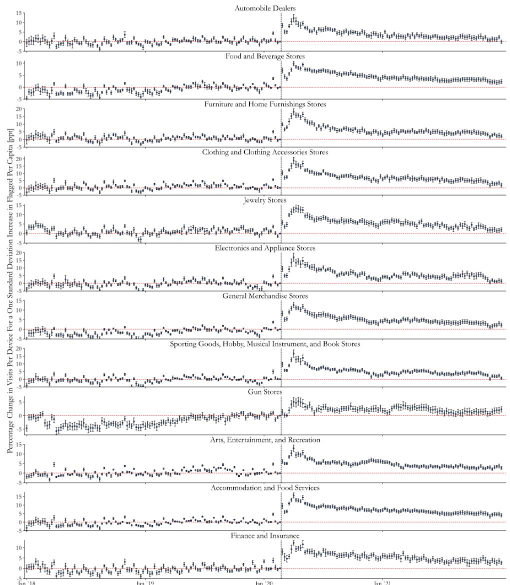


Fig. 7. Effect on consumer visits to various types of locations. This figure shows the effect of suspicious lending on consumer visits to various types of locations. Data on consumer visits is from SafeGraph and has been aggregated to the census tract × week × location type level. Location types are defined by NAICS codes; gun stores are identified using a combination of NAICS codes and a keyword search of location names. Each subpanel examines the effects of a one standard deviation change in the number of flagged PPP loans per capita on visits to the given type of location. The dependent variable is $IHS(Visits)_{itl} - IHS(Devices)_{it}$, where $IHS(.)$ is the inverse hyperbolic sine function, i is a census tract, t is a week, and l is a type of location. Note that the coefficients can be interpreted as a percentage change by the same logic as when the dependent variable is log transformed. The regressions include census tract fixed effects, week × county fixed effects, and control for post × loans per capita. Both the independent and dependent variables are winsorized at the 1% tails. The week of February 17th to 24th, 2020 is the omitted week. The error bars represent 95% confidence intervals based on standard errors that are double clustered by county and week. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the census tract's population as of 2019.

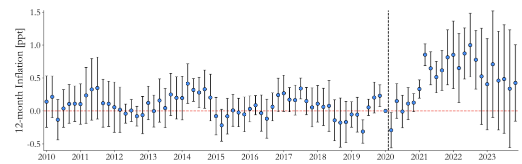


Fig. 8. Effect on Regional Inflation. This figure shows the effect of suspicious lending on regional inflation using regional all-items consumer price indices (CPI) from the BLS. Data for 23 CBSAs is released bi-monthly. 12-month inflation is determined by dividing the given month's CPI by the CPI 12 months earlier. *Flagged Per Capita* is standardized, so the coefficients represent the inflation effect of a one standard deviation change in suspicious lending. To have a nationally representative estimate, we use weighted least squares (WLS) regressions with the weight being the CBSA's population as of 2019. The error bars represent 95% confidence intervals based on standard errors that are double clustered by CBSA and bi-month.

5.18 Figure 8: Regional inflation

Read:

- Before COVID, regional inflation has little relation to future local fraud intensity.
- Beginning in late 2020, inflation in high-fraud CBSAs rises relative to low-fraud CBSAs, peaking in mid-2022.
- The figure connects the micro-spending results to a macro price consequence: local fraud appears to have contributed to regional inflation differences.

5.19 Table 11: Regional inflation

Read:

- The post-period coefficient on flagged fraud is 0.429 ppt for overall inflation.
- For the housing component of CPI, the coefficient is 0.707 ppt, which is even larger.
- For all-items excluding shelter, the effect is small and only significant in one specification, at 0.252 ppt.
- The table fits the paper's main mechanism: fraud creates the strongest inflationary effect where local supply is least flexible, namely housing.

6 Short summary

- **Conclusion.** Pandemic-relief fraud was not just a transfer from taxpayers to fraudsters; it also appears to have temporarily inflated house prices and local spending in the places where it was concentrated.
- **Intuition.** Fraud created a windfall, recipients spent part of that windfall on houses, vehicles, and consumption, and local prices moved because the spending shock was geographically concentrated.
- **Empirical lesson.** Comparing suspicious and non-suspicious relief flows is a powerful way to separate true stimulus from income replacement. The paper shows that this distinction matters enormously for understanding the post-COVID housing boom.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Fraudulent pandemic-relief funds stimulated local purchases and pushed up local house prices.
- **Magnitude.** The effect is economically large: 5.8 ppt more house-price growth in top-decile versus bottom-decile fraud ZIP codes within the same county, 2.11 ppt for a one-standard-deviation increase in flagged fraud in the baseline regression, and around 3.47 ppt in the IV design.
- **Micro foundation.** Recipients of suspicious loans are more likely to buy homes and move, so the price effect comes from actual behavior rather than only reduced-form geography.
- **Mechanism evidence.** High-fraud places become tighter housing markets in 2020–21 and looser markets in 2022–23, exactly as a temporary local demand shock would predict.
- **Broader spending effects.** Fraud-heavy places also show stronger vehicle purchases, higher consumer spending, more retail visits, and more regional inflation.
- **Persistence and reversal.** The house-price effect is not permanent. Once the spending wave passes, a meaningful share of the initial distortion unwinds.

7.2 Economic intuition

- Legitimate relief transfers mainly replaced lost income or revenue; fraud did not. That difference is the core conceptual insight of the paper.
- Because fraud was geographically concentrated, even spending by a relatively small number of recipients could create a sizable local demand shock.
- Housing is especially sensitive because it is a local, immovable asset with inelastic short-run supply. That is why the strongest spending distortion appears in house prices and housing inflation.

- The later reversal is just as informative as the initial boom. It implies the effect was not simply better local fundamentals, but a temporary distortion tied to a temporary wave of fraudulent cash.

7.3 Takeaway

This paper shows that the external costs of fraud can extend far beyond the stolen dollars themselves. Pandemic-relief fraud appears to have reallocated purchasing power into particular neighborhoods, temporarily raising house prices and broader local spending. The broader lesson is policy-relevant: when governments design large emergency programs without strong fraud controls, the resulting distortions may spill into asset markets, consumer markets, and inflation in ways that are both large and persistent enough to matter for macroeconomic outcomes.